

CS 662: Mobile Virtual Reality and Artificial Intelligence

Course Code	CS 662
Course Name	Mobile Virtual Reality and Artificial Intelligence
L-T-P-C	3-0-0-3
Prerequisites	CS 660, CS 669, or an equivalent course in AI/ML area
Intended for	B. Tech./M. Tech ./M.S./Ph.D. students
Approval	recommended by the 37th BoA (Oct 2020)

Course Contents

- **Introduction to VR and VR programming** [12 Lectures]
- **Building the first AI simulation** :Developing a Pathfinding Game, How to Set Up a Project, Node, String Map, A* Algorithm Setup, A* Algorithm Loop, Auxiliary Methods, Finishing the Algorithm, Importing 2D Assets, Building a Level, From Console to Visual, Adding Tanks, Identifying Nodes, Moving the Tank, Visually Moving Tank, Smooth Movement, Smooth Rotation, Ordering Tank to Move, Speeding up Player, Spawning Logic, Crate Visuals, Adding Crates to Valid Positions, Collecting Crates, Score Counting, Game Interface, Starting the Game, Game Over Screen, Scoring, Sounds. [6 Lectures]
- **Introduction to Virtual Reality and Blending**: VR Introduction, Camera Changing Position, Triggering Events Interface, Blending and Introduction & Customizing Settings, Controlling Blender Camera, Emulate Numpad Camera, Manipulating Objects, Common Tools, Mirroring 1 Side of Object. Case Study: Flappy bird game in VR, First person shooter VR game, Kart VR game. [8 Lectures]
- **Introduction to ML in VR**: Introduction to Machine Learning, the need for ML in VR, different kinds of learn ings, Neural Networks (NNs), Training a NN, Optimizer, Convolutional layers, Transfer learning, Imitation learning. Case study: Training the karin kart game via IL and testing the kart drive.
- **Reinforcement Learning**: Introduction to Reinforcement Learning, Initial state, Training a policy, The PPO algorithm, Evolutional Strategies, Reward. Case study: Training a kart in the kart game with RL, Tensorboard analysis, Testing results. [10 Lectures]

Text Books

1. Buttfield-Addison, P., Manning, J., Nugent, T., **Unity Game Development Cookbook: Essentials for Every Game**, O'Reilly Media, 2019.

References

1. Linowes, J., & Schoen, M, **Cardboard VR Projects for Android**, Packt Publishing, 2016.
2. Lanham, M., **Hands-On Deep Learning for Games: Leverage the power of neural networks and reinforcement to build intelligent games**, Packt Publishing Ltd, 2019.